NetPractice: Full Guide to 100%

42 School study guide: concepts, a solving method, practice with answers, and evaluation prep.

Part Z: Start here, everything from zero

This part assumes you know nothing about networking. Every new word is explained the first time it appears, and every number is explained: where it comes from and why it is used. Parts 1 to 10 further down are the compact version of the same material, so after you finish Part Z they will feel easy. If a sentence is still unclear, comment on it and I will rewrite it.

Z1. What problem are we solving?

Computers need to send messages to each other: your laptop asks a website for a page, your phone sends a photo to a friend. A **network** is a group of devices connected so they can send each other messages. The connection is a cable (Ethernet) or radio (Wi-Fi).

The problem: if 100 devices are connected, how does a message reach the right one? Every device gets an **address**, and every message carries the address of its destination. It works like posting a letter: you write the address on the envelope and the postal system delivers it.

I will use a city comparison through the whole guide:

In networking	In a city
Device	A house
IP address	The full address of that house
Network (also called subnet)	One street
Switch	The road that links all the houses on one street
Router	A crossroads that belongs to two or more streets at once
Gateway	The exit of your street toward other streets
Packet	One letter

Quick definitions:

- **Device**: anything that can send or receive messages: a PC, phone, server, printer, router.
- **Host**: an ordinary end device such as a PC or a server (not a router or a switch). In NetPractice the hosts are the computers you must connect.
- **Packet**: a small piece of data with the destination address attached. Everything you do online is cut into packets.
- **Protocol**: an agreed set of rules, like a shared language, so devices from different makers understand each other.

NetPractice in one sentence: the website shows you a drawing of devices and asks you to fill in addresses and rules so that every device that should be able to talk to another one can.

Z2. Why numbers like 255, 256 and 192? Binary from zero

People count in base 10: ten digits (0 to 9), probably because we have ten fingers. Computers only have two states (electricity on or off), so they count in **base 2**, called **binary**, which has only two digits: 0 and 1.

- A **bit** is one binary digit (a single 0 or 1).
- A **byte** is 8 bits. In networking people say **octet** ('octo' means eight) for the same thing.

How base 10 works: 345 means 3 x 100 + 4 x 10 + 5 x 1. Each position is worth 10 times the position on its right.

How base 2 works: each position is worth 2 times the position on its right. 1, 2, 4, 8, 16, 32, 64, 128. A 1 means 'count this value' and a 0 means 'skip it'.

```
position value:  128   64   32   16    8    4    2    1
bits:              1    0    1    0    1    1    0    0
counted:         128 +  0 + 32 +  0 +  8 +  4 +  0 +  0  = 172
```

So 10101100 in binary is 172 in decimal.

Where do 255 and 256 come from? With 8 bits, the biggest number is all ones: 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255, and the smallest is all zeros = 0. So one octet can hold 256 different values (0 to 255). Every time you see 255 or 256 in this guide, it is this: 256 values, from 0 to 255.

Powers of 2: 2^n means 2 multiplied by itself n times. 2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 32, 2^6 = 64, 2^7 = 128, 2^8 = 256. These are the same numbers as the position values above (plus 256). Learn this row by heart, it is used everywhere.

Decimal to binary with the subtract method. Convert 130: go through the positions from the left. Does 128 fit in 130? Yes, write 1, remainder 2. Does 64 fit in 2? No, write 0. Then 32, 16, 8 and 4 do not fit either: 0, 0, 0, 0. Does 2 fit in 2? Yes, write 1, remainder 0. Does 1 fit in 0? No, write 0. Result: 10000010.

Now the numbers you will see again and again in masks. Each one is the previous one plus the next position value:

Decimal	Binary	How it is built
128	10000000	the first position only
192	11000000	128 + 64
224	11100000	192 + 32
240	11110000	224 + 16
248	11111000	240 + 8
252	11111100	248 + 4
254	11111110	252 + 2
255	11111111	254 + 1

That is the whole reason masks use the values 128, 192, 224, 240, 248, 252, 254 and 255: they are 1, 2, 3, up to 8 ones counted from the left.

Try it, then check the answers. Convert 200, 77 and 10 to binary, and convert 11000101 to decimal.

Answers: 200 = 128 + 64 + 8 = 11001000. 77 = 64 + 8 + 4 + 1 = 01001101. 10 = 8 + 2 = 00001010. 11000101 = 128 + 64 + 4 + 1 = 197.

Z3. What is an IP address?

IP stands for Internet Protocol. An **IP address** is the address of one network connection of one device. An IPv4 address (the 'v4' means version 4) is 32 bits long, cut into 4 octets (4 x 8 = 32). To make it readable, each octet is written as a normal decimal number and they are separated by dots. This is called **dotted decimal**:

```
11000000.10101000.00000001.10000010   what the computer sees (32 bits)
   192   .   168   .    1    .  130    what humans read
```

Each part is between 0 and 255 because each part is one octet (see Z2).

How many IPv4 addresses exist? 2^32 = 4,294,967,296, about 4.3 billion. That is no longer enough for the whole world, so a newer version, IPv6 (128 bits), exists. NetPractice only uses IPv4.

An **interface** is one connection point of a device to a network: a Wi-Fi card, a cable socket, one port of a router. IP addresses belong to interfaces, not to whole devices. A laptop using Wi-Fi and a cable has two interfaces and so two IP addresses. A router connected to three networks has three interfaces with three different IP addresses. This matters a lot in NetPractice because routers have many.

Every IP address has two parts, like a postal address has a street and a house number:

- the **network part** says which street you are on;
- the **host part** says which house on that street.

All devices on the same street have the same network part and a different host part. For example 192.168.1.10 and 192.168.1.20 can be neighbours: same street (192.168.1), houses 10 and 20. But the place where the street name ends is not always at a dot. The **subnet mask** tells you exactly where, and that is the next section.

Z4. The subnet mask (the most important idea)

Problem: how does a device know where the network part ends and the host part starts? Answer: the **subnet mask**, which is also 32 bits. The rule is simple. Write the mask in binary. Every bit that is 1 means 'this bit of the address is part of the network'. Every bit that is 0 means 'this bit is part of the host'. A mask is always a row of 1s followed by a row of 0s, never mixed.

Example: 255.255.255.0 is 11111111.11111111.11111111.00000000. That is 24 ones then 8 zeros, so the first 24 bits are the network and the last 8 bits are the host. With the address 192.168.1.10, the network part is 192.168.1 and the host part is 10.

CIDR notation (say 'cider') is a shorthand: after the address you write a slash and the **number of ones** in the mask. 255.255.255.0 has 24 ones, so it is written /24. 255.255.0.0 is /16 and 255.0.0.0 is /8. The number after the slash is also called the **prefix length**. (CIDR stands for Classless Inter-Domain Routing; you only need the meaning of the slash.)

Where does 255.255.255.192 come from for /26? Three full octets of ones are 24 ones. We need 2 more ones, which go in the 4th octet: 11000000 = 192. So /26 = 255.255.255.192. The same logic gives every row:

Prefix	Ones in the 4th octet	4th octet in binary	Mask
/24	0	00000000	255.255.255.0
/25	1	10000000	255.255.255.128
/26	2	11000000	255.255.255.192
/27	3	11100000	255.255.255.224
/28	4	11110000	255.255.255.240
/29	5	11111000	255.255.255.248
/30	6	11111100	255.255.255.252

For prefixes between /16 and /24 the extra ones fall in the 3rd octet. For example /20 = 255.255.240.0, because 16 ones fill the first two octets and 4 more ones in the 3rd octet make 11110000 = 240.

AND is the operation a device uses to apply the mask. It compares two bits and gives 1 only if both are 1: 1 AND 1 = 1, 1 AND 0 = 0, 0 AND 1 = 0, 0 AND 0 = 0. Think of the mask as a stencil: where the mask has a 1, the address bit shows through. Where the mask has a 0,

the result is forced to 0. The result of IP AND mask is the **network address**: the name of the street.

Full example: 192.168.1.130 with /26.

```
IP:      11000000.10101000.00000001.10000010   192.168.1.130
Mask:    11111111.11111111.11111111.11000000   255.255.255.192 (/26)
AND:     11000000.10101000.00000001.10000000   192.168.1.128
```

The first three octets of the mask are all ones, so those octets pass through unchanged. In the last octet, 10000010 AND 11000000 = 10000000, which is 128. So the network address is 192.168.1.128.

Two devices are on the same network (same street) if IP AND mask gives the same result for both. If the results differ, they are on different streets and need a router between them.

Z5. Network address, broadcast address, and why 'minus 2'

Inside one network, the **host bits** are the bits under the zeros of the mask. Call their count h, where h = 32 minus the prefix. With h bits you can write 2^h different patterns, so the network has 2^h addresses. Two of them are special and can never be given to a device:

- All host bits 0 = the **network address**. It is the name of the street itself, not a house.
- All host bits 1 = the **broadcast address**. A message sent to it is delivered to every device on the street.

Two more words: **unicast** means a message to one device, and **broadcast** means a message to all devices of the network.

So the number of usable addresses is 2^h - 2. That is the 'minus 2' you keep seeing.

Example with /24: h = 32 - 24 = 8, and 2^8 = 256 addresses, from 192.168.1.0 to 192.168.1.255. The address .0 is the network address and .255 is the broadcast, so the usable ones are .1 to .254, which is 254 addresses. That is where the number 254 comes from: 256 - 2.

Prefix	Host bits h (32 - prefix)	Addresses 2^h	Usable 2^h - 2
/30	2	4	2
/29	3	8	6

Prefix	Host bits h (32 - prefix)	Addresses 2^h	Usable 2^h - 2
/28	4	16	14
/27	5	32	30
/26	6	64	62
/25	7	128	126
/24	8	256	254
/23	9	512	510
/16	16	65,536	65,534

See it in binary with the network 192.168.1.128/26. The last octet is split like this: the first 2 bits are network bits (they are '10'), the last 6 bits are host bits.

```
network address:   10 000000  = 128   (host bits all 0)
first usable:      10 000001  = 129
last usable:       10 111110  = 190
broadcast:         10 111111  = 191   (host bits all 1 = 128 + 63)
```

The 63 is 2^6 - 1: six host bits all set to 1 add up to 32 + 16 + 8 + 4 + 2 + 1 = 63. That is why this subnet ends at 191.

Z6. The block-size trick, and why it works

Do not memorize '256 minus the mask value' blindly. Here is why it works.

Take /26. The last octet of the mask is 192 = 11000000. It has 6 zeros, so there are 6 host bits, so each subnet has 2^6 = 64 addresses. And 256 - 192 = 64. It is the same number, and this is why: if the octet has k zeros at the end, the mask value is 256 - 2^k (all ones except the last k bits), so 256 minus the mask value gives back 2^k, the number of addresses.

Mask octet	Zeros k	Block size 2^k
128	7	128
192	6	64
224	5	32

Mask octet	Zeros k	Block size 2^k
240	4	16
248	3	8
252	2	4

The **block size** is the number of addresses in one subnet, counted in the 'interesting octet' (the octet where the mask is not 255 and not 0). The 256 values of that octet are cut into equal pieces and each piece is one subnet. For /26:

```
0 ........ 63 | 64 ........ 127 | 128 ........ 191 | 192 ........ 255
 network  broadcast  network  broadcast  network  broadcast  network
broadcast
```

The first number of each piece is its network address, the last number of each piece is its broadcast address, and everything between is usable. Pieces always start at multiples of the block size (0, 64, 128, 192) because every piece is exactly 64 long and the first one starts at 0.

To find the subnet of any address, ask: which piece does it fall into? For 192.168.1.130 the piece is 128 to 191. So the network is 192.168.1.128, the broadcast is 192.168.1.191, the usable range is .129 to .190, and there are 62 hosts.

Another example, 10.1.1.77/27. For /27 the mask octet is 224, so the block is 256 - 224 = 32. The pieces are 0-31, 32-63, 64-95, 96-127, and so on. 77 falls in 64-95. Network 10.1.1.64, broadcast 10.1.1.95, usable .65 to .94, which is 30 hosts (2^5 - 2).

When the mask boundary is in the third octet (prefixes /17 to /23) you use the same trick on the third octet, and the whole fourth octet is host bits (0 to 255). Example: 172.16.37.5/20. The mask is 255.255.240.0, so the interesting octet is the third one and the block is 256 - 240 = 16. The pieces are 0-15, 16-31, 32-47. The number 37 falls in 32-47, so the network is 172.16.32.0. The broadcast has third octet 47 and a fourth octet of all ones, which is 255: 172.16.47.255. The usable range is 172.16.32.1 to 172.16.47.254. The host bits are h = 32 - 20 = 12, and 2^12 - 2 = 4094 hosts.

Notice that inside this subnet, 172.16.32.255 and 172.16.33.0 are perfectly normal hosts. The only forbidden addresses are the all-zero and all-one host patterns, which are 172.16.32.0 and 172.16.47.255. The same idea explains why 10.0.0.255 is a broadcast in 10.0.0.0/24 but a normal host in 10.0.0.0/23, where the range is 10.0.0.0 to 10.0.1.255.

Z7. How many addresses do I need? (sizing a subnet)

Steps: (1) count every device that needs an address in that subnet, including every router interface; (2) add 2 for the network and broadcast addresses; (3) find the smallest power of 2 that is at least that big the exponent is h; (4) the prefix is 32 - h.

- 50 PCs + 1 router interface = 51 devices. 51 + 2 = 53. The smallest power of 2 that is 53 or more is 64 = 2^6, so h = 6 and the prefix is 32 - 6 = /26. Check: 62 usable is at least 51. Why not /27? That has only 32 addresses and 30 usable, which is less than 51.
- 20 PCs + 1 router = 21. 21 + 2 = 23. The next power of 2 is 32 = 2^5, so /27 (30 usable).
- 500 hosts: 500 + 2 = 502. The next power of 2 is 512 = 2^9, so /23 (510 usable).
- A link between two routers: 2 interfaces + 2 reserved = 4 = 2^2, so /30. A /31 has only 2 addresses and so no room for the reserved two in the classic rules, and a /32 is a single address. NetPractice follows the classic rule, so /30 is the smallest subnet you use.

Alignment rule: a subnet of size 2^h must start at a multiple of 2^h, because the pieces line up like in Z6. A /27 (32 addresses) can start at .0, .32, .64, .96 and so on, but not at .50. In the practice answer F1, the /26 starts at .0 and the /27 starts at .64, which is a multiple of 32, so it is valid.

Why choose tight sizes? Some levels give you one range and ask for several subnets inside it. If you waste addresses the pieces no longer fit. For example inside 192.168.1.0/24 (256 addresses), a /26 (64) plus a /27 (32) use 96, so there is plenty of room.

Z8. Same street or not? The decision every device makes

A host that wants to send a packet does this:

1. Take my own IP and my own mask, apply AND. The result is my network.
2. Take the destination IP, apply my own mask with AND. The result is the destination's network as I see it.
3. If the two results are equal, the destination is on my street, so I send it directly. If not, I send it to my default gateway.

Example: host 10.0.1.10 with /24 wants to reach 10.0.2.10. 10.0.1.10 AND 255.255.255.0 = 10.0.1.0 (my network). 10.0.2.10 AND 255.255.255.0 = 10.0.2.0. They are different, so the packet goes to the gateway.

Why the masks must be identical on one street: imagine A = 192.168.1.10/24 and B = 192.168.1.130/25 on the same switch. A computes its own network 192.168.1.0 and the network of B's address under /24, also 192.168.1.0, so A thinks B is local and sends directly.

B computes its own network under /25: 130 AND 128 = 128, so 192.168.1.128. For A's address it computes 10 AND 128 = 0, so 192.168.1.0, which is different from its own, so B thinks A is remote and sends the reply to its gateway. The conversation breaks because the two devices disagree about where the street ends.

Why the gateway must be on your own street: a host can only deliver directly to devices on its own street, and reaching the gateway is also a direct delivery. A gateway that is not on your street is unreachable, so it is useless.

Z9. The devices, in detail

First the words:

- **LAN** (Local Area Network): one local network, like a home or a school lab. **WAN** (Wide Area Network): links between far-away places. The **Internet** is the giant network of networks joined together all over the world.
- **MAC address**: a 48-bit hardware address burned into every network card, written in hexadecimal like 00:1A:2B:3C:4D:5E. An IP address is configured and can change; a MAC address is like a serial number. MAC addresses work only inside one street (one local link).
- **Switch**: a box with many ports. It learns which MAC address sits behind which port and forwards each message only to the right port. In NetPractice a switch has no IP address and does not read IPs or masks. Everything plugged into one switch is one street. So in a drawing, a switch means 'these devices must be in the same network with the same mask'. (An old device called a hub just copies every message to every port.)
- **Router**: a switch only joins devices of the same street. To pass from one street to another you need a device that belongs to both. A router has one interface (one 'leg') on each street, and each leg has an IP address from that street. It forwards packets from one interface to another using its routing table. Why must a router's interfaces be in different networks? The router chooses the leg by looking at the network of the destination. If two legs were on the same street it could not tell them apart.
- **Gateway** (or **default gateway**): the IP address of the router interface on your own street that you send everything to when the destination is not on your street. 'Default' because it is the default exit.
- **ARP** (Address Resolution Protocol): on a street, delivery uses MAC addresses, but you only know the IP. ARP shouts to the whole street: 'who has IP 10.0.1.1? tell me your MAC'. The owner answers, and then the packet is delivered to that MAC.

- **Frame**: local delivery needs an envelope with MAC addresses on it. A packet inside that envelope is called a frame. At each router the old envelope is thrown away and a new one is put on for the next street. The packet inside, with its source and destination IP, stays the same.
- **TTL** (Time To Live): a number inside every packet (for example 64). Each router lowers it by 1. At 0 the packet is destroyed. This protects the network from packets bouncing forever in a loop.
- **Ping**: a test tool. It sends a tiny 'are you there?' message to an IP and waits for the answer. The Check again button in NetPractice is basically testing that devices can reach what they should, and that the replies come back.
- **Layers**: networking is built in layers. Layer 2 (link) is local delivery by MAC addresses, which is the switch's job. Layer 3 (network) is delivery between networks by IP address, which is the router's job. That is the short answer to 'switch vs router'.

Z10. Routing, from zero

Routing means deciding where to send a packet next. A router keeps a **routing table**, which is a list of rules. Each rule is called a **route**. A route says: 'for destinations in network X, hand the packet to neighbour Y'. A route has two columns:

- **Destination**: a whole network, written as the network address plus the prefix, like 10.10.2.0/24.
- **Next hop**: the IP address of the next router's interface, which has to be a direct neighbour.

Networks the router is directly connected to need no rule, because the router is already standing on those streets.

You write the rules by hand, which is called a **static route**. Big real networks learn routes automatically with dynamic routing protocols (OSPF, BGP). You do not need those here.

Why must the next hop be directly reachable? The router hands the packet to a neighbour over a street it is on. A next hop two streets away cannot be delivered to directly, which is the same logic as the gateway rule in Z8.

Why must the destination be a network address? The rule describes a whole street, not one house. 10.10.2.5/24 mixes a house with a street. The correct form is 10.10.2.0/24, with all host bits at 0.

Longest prefix match. If several rules match the same destination, the router uses the one with the longest prefix, because more matching network bits means more specific. It is like preferring 'house 5 on Main Street' to 'somewhere in the city'. Example: rules 10.0.0.0/8 via X and 10.1.0.0/16 via Y. A packet for 10.1.2.3 matches both (8 bits and 16 bits match). 16 is longer than 8, so it goes via Y. A packet for 10.9.9.9 matches only the /8, so it goes via X.

Default route, 0.0.0.0/0. A prefix of /0 means zero network bits have to match, so every address matches. That is why it means 'everything else'. It loses against every more specific rule because its prefix is the shortest possible. The address part is 0.0.0.0 by convention (any value would do, since no bits are compared). You use it for 'send everything I do not know toward the exit or the Internet'. Some interfaces write it as default, which means the same.

Both directions matter. A ping needs a request and a reply. If R2 has no route back to LAN A, the request arrives but the reply is lost, and the test fails.

Loops: if R1's default route points to R2 and R2's default route points back to R1, a packet for an unknown network bounces between them until its TTL reaches 0. Default routes must point toward the exit, never back.

Worked example with reasons for every number:

```
LAN A 10.10.1.0/24 -- R1 ==(10.10.3.0/30)== R2 -- LAN B 10.10.2.0/24
```

- Three separate streets: LAN A, LAN B, and the cable between R1 and R2. So three subnets.
- The two LANs are /24 because the exercise gives them (254 hosts each).
- The link is /30 because it needs 2 router interfaces + 2 reserved = 4 addresses (Z7).
- Why 10.10.3.0 for the link? Any free number works. 10.10.1.x and 10.10.2.x are already used, so I took the next third octet. It must not overlap the other two.
- In a /30 the usable addresses are only .1 and .2, so R1 gets 10.10.3.1 and R2 gets 10.10.3.2.
- Each router gets the first usable address on its LAN by habit: R1 = 10.10.1.1/24, R2 = 10.10.2.1/24. Hosts then use .2 and up.
- Hosts use as gateway the router interface on their own street: 10.10.1.1 for LAN A, 10.10.2.1 for LAN B.
- R1 is directly connected to LAN A and the link, but does not know LAN B. So R1 needs: 10.10.2.0/24 via 10.10.3.2 (R2 is its neighbour on the link).

- R2 is directly connected to LAN B and the link, but does not know LAN A. So R2 needs: 10.10.1.0/24 via 10.10.3.1.

Trace of a ping from a host in A to a host in B: the host sees B is not local and sends to 10.10.1.1 (R1). R1 looks in its table: 10.10.2.0/24 via 10.10.3.2, so it forwards to R2. R2 sees that 10.10.2.0/24 is directly connected and delivers to the host. The reply travels back the same way using R2's route to 10.10.1.0/24.

Z11. Private vs public addresses, and why 10, 172.16 and 192.168

A **public** address is unique in the whole world and is assigned through internet providers. A **private** address is reserved for use inside networks. Routers on the Internet refuse to forward private addresses, so every home and school can reuse the same ones. They were defined in a document called RFC 1918 (an RFC, 'Request For Comments', is one of the numbered documents that define Internet rules).

Range	Prefix	From	To	Addresses
10.0.0.0/8	8	10.0.0.0	10.255.255.255	16,777,216 (2^24)
172.16.0.0/12	12	172.16.0.0	172.31.255.255	1,048,576 (2^20)
192.168.0.0/16	16	192.168.0.0	192.168.255.255	65,536 (2^16)

Why does the middle range stop at 172.31? A /12 means the first 12 bits are fixed: the whole first octet (172, which is 8 bits) plus the first 4 bits of the second octet. The number 16 in binary is 00010000, so those 4 fixed bits are 0001. The other 4 bits of the second octet can be anything from 0000 to 1111, which adds 0 to 15. So the second octet runs from 16 to 31. The address counts come from 2^(32 - prefix): 2^24, 2^20 and 2^16.

NAT (Network Address Translation): the router at the edge replaces the private source address of outgoing packets with its own public address, and remembers how to translate the replies back. That is how a house full of private devices shares one public IP. NetPractice does not simulate NAT, but an evaluator may ask about it.

Other special addresses:

- 127.0.0.0/8: loopback. 127.0.0.1 is 'localhost', meaning yourself. It never leaves your machine.
- 169.254.0.0/16: link-local. A device gives itself one of these when it cannot get an address.

- 0.0.0.0: means 'unspecified' or 'any'. It is also the address part of the default route 0.0.0.0/0.
- 255.255.255.255: broadcast to everyone on the local street.
- **DHCP**: a service that hands out addresses automatically (your home router does it). In NetPractice you set addresses by hand, which is called static.

In NetPractice, use private ranges for your own internal networks, and do not reuse the Internet's fixed public addresses for your hosts.

Z12. The numbers in the practice answers, explained step by step

The practice answers in Part 8 are short. Here is where their numbers come from.

- A1, 192.168.10.200/26: the mask octet is 192, so the block is 256 - 192 = 64. The pieces are 0-63, 64-127, 128-191, 192-255. The number 200 falls in 192-255. The network is 192.168.10.192. The broadcast is 192 + 64 - 1 = 255. The usable range is .193 to .254, and 62 hosts because 2^6 - 2 = 62.
- A5, 10.200.77.5/18: /18 means 16 ones fill the first two octets and 2 more ones are in the third octet: 11000000 = 192, so the mask is 255.255.192.0. The block in the third octet is 64. The number 77 falls in 64-127. Network 10.200.64.0, broadcast 10.200.127.255 (third octet 127, fourth octet all ones = 255). Host bits 32 - 18 = 14, and 2^14 = 16,384, minus 2 = 16,382 hosts.
- B3, the /23 case: /23 means 16 ones + 7 ones in the third octet: 11111110 = 254, so the mask is 255.255.254.0. The block is 256 - 254 = 2, so the pieces in the third octet are 0-1, 2-3, 4-5, and so on. For 172.16.5.1 the third octet is 5, which is in the piece 4-5. For 172.16.4.200 the third octet is 4, which is in the same piece. Both are in the network 172.16.4.0/23, which runs from 172.16.4.0 to 172.16.5.255. So they are on the same street.
- C1, 192.168.5.64/27: 64 is a multiple of 32, so it starts a piece, which makes .64 the network address. The broadcast is 64 + 32 - 1 = 95. The address .96 starts the next subnet. So only .70 is a valid host.
- E1, host 10.0.0.70/26 with gateway 10.0.0.1: the block is 64, so the pieces are 0-63 and 64-127. The host (70) is in 64-127, so its network is 10.0.0.64. The gateway 10.0.0.1 is in 0-63, which is a different street, so the host cannot reach it. A valid gateway would be 10.0.0.65.
- F1: the sizes come from Z7 (51 addresses needed gives /26, 21 gives /27). The router gets the first usable address of each subnet by habit: .1 for 192.168.1.0/26 and .65 for 192.168.1.64/27 (64 + 1). Hosts use the addresses after that.

- F3: 10.0.12.0/30 and 10.0.23.0/30 are just free private networks, and the numbers 12 and 23 only help me remember 'between router 1 and 2' and 'between router 2 and 3'. Any free /30 works. The address 203.0.113.0/30 comes from a block that is reserved for examples in documentation, so it is a safe way to pretend 'this is a public address'.

Z13. Glossary

Word	Meaning
AND	Bit operation: 1 only if both bits are 1. Used to apply a mask to an IP address.
ARP	Protocol that finds the MAC address that belongs to an IP address on the local street.
Bit	One binary digit, 0 or 1.
Block size	Number of addresses in one subnet, counted in the interesting octet: 256 minus the mask octet.
Broadcast address	Last address of a subnet (all host bits 1). A message sent there reaches every device on the street.
Byte	8 bits. Same as an octet.
CIDR / prefix length	The slash number, which is the count of 1s in the mask. /24 = 255.255.255.0.
Default gateway	The router interface on your own street used as the exit toward other networks.
Default route	The route 0.0.0.0/0 that matches everything. Used when no more specific route exists.
Destination	In a route, the network the rule is about, like 10.10.2.0/24.
DHCP	A service that gives devices their addresses automatically.
Dotted decimal	Writing an IP as four decimal numbers separated by dots.

Word	Meaning
Frame	A packet inside a local envelope with MAC addresses.
Host	An end device such as a PC or server.
Host bits	The bits of an address under the zeros of the mask. They identify the device.
Interface	One network connection point of a device. It carries one IP address.
Internet	The worldwide network of networks.
IP address	32-bit address of an interface (IPv4).
LAN	Local Area Network, one local network.
Loopback	127.0.0.1, the address of yourself.
MAC address	Hardware address of a network card, used for delivery inside one street.
Mask	32-bit number whose 1s mark the network part of an address.
NAT	The router replaces private addresses with its public one when leaving to the Internet.
Network address	First address of a subnet (all host bits 0). It names the street.
Network bits	The bits of an address under the ones of the mask. They identify the street.
Next hop	In a route, the IP of the neighbouring router interface to give the packet to.
Octet	8 bits, one of the four parts of an IPv4 address, value 0 to 255.
Packet	A piece of data with source and destination addresses.
Ping	A test message that checks if a device answers.

Word	Meaning
Private address	Address reserved for inside networks (10/8, 172.16/12, 192.168/16).
Protocol	An agreed set of rules for communication.
Public address	Address that is unique worldwide and reachable on the Internet.
Route	One rule in a routing table: destination network plus next hop.
Router	Device with several interfaces on different networks that forwards packets between them.
Routing table	The list of routes of a router.
Subnet	One network (one street) cut out of a bigger address range.
Switch	Layer 2 device that connects devices of the same network. It has no IP in NetPractice.
TTL	Counter in every packet, lowered by 1 per router, packet dies at 0.
Unicast	A message to one single device.

Z14. Self-test and study order

Answer these on paper, then check.

1. Why does an octet stop at 255?
2. Write 224 in binary, and say which prefix you get when the first three octets are 255 and the fourth is 224.
3. Find the network, broadcast, usable range and host count of 172.31.200.77/21.
4. How many usable hosts are in a /28?
5. Why must a host's gateway be on the host's own street?
6. What does 0.0.0.0/0 mean, and why does it match everything?
7. Can 192.168.1.64/26 be given to a host?
8. A link between two routers: how many addresses are needed and which prefix?
9. Why can a ping fail even if the route toward the destination is correct?

10. Which are private: 172.15.0.1, 172.20.0.1, 192.169.0.1?

Answers:

1. One octet is 8 bits. All ones = 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255.
2. 224 = 11100000, which is 3 ones, so the prefix is 24 + 3 = /27.
3. /21 = 16 ones + 5 ones in the third octet = 11111000 = 248. Block = 256 - 248 = 8. The third octet 200 is a multiple of 8 (8 x 25), so the piece is 200-207. Network 172.31.200.0, broadcast 172.31.207.255, usable 172.31.200.1 to 172.31.207.254, host bits 32 - 21 = 11, 2^11 - 2 = 2046 hosts.
4. 2^4 - 2 = 14.
5. A host only delivers directly to its own street, and reaching the gateway is a direct delivery.
6. A prefix of /0 means no bits have to match, so every address matches. It has the shortest prefix, so any more specific route wins over it.
7. No. 64 is a multiple of 64, so it is the network address of 192.168.1.64/26.
8. 2 interfaces + 2 reserved = 4 addresses, so a /30.
9. The reply needs a route back. If the return route or the other side's gateway is wrong, the reply is lost.
10. Only 172.20.0.1. The private range is 172.16.0.0 to 172.31.255.255, and 192.168 (not 192.169) is the private one.

Suggested study order: day 1, Z1 to Z4 and the exercises in Z2. Day 2, Z5 to Z8 and answers A1 to A6 of Part 8. Day 3, Z9 to Z11, then Part 8 B to F. Then solve the levels with the method in Part 7, and use Part 9 to rehearse the evaluation out loud.

0. How to use this guide

Start with Part Z, which explains everything from zero. Then read parts 1 to 6 once as the compact version. Do the practice in part 8 on paper, with no calculator. Then solve the levels using the method in part 7. Part 9 is your evaluation prep. The subject PDF and the eval scale are the final authority, so reread both before you start and again before the defense. Where this guide mentions 42-specific details, treat them as typical and verify them against your subject.

1. Numbers to know by heart

Powers of 2: 1, 2, 4, 8, 16, 32, 64, 128, 256.

Mask octet values (1 to 8 ones): 128, 192, 224, 240, 248, 252, 254, 255.

CIDR	Mask	Block size (interesting octet)	Usable hosts
/8	255.0.0.0	256 in 1st octet	16,777,214
/16	255.255.0.0	256 in 2nd octet	65,534
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

Formulas: usable hosts = 2^(32 - n) - 2. Block size = 256 - mask octet value.

2. IP addresses and binary

An IPv4 address is 32 bits, written as 4 octets of 0 to 255. Each octet is 8 bits with values 128, 64, 32, 16, 8, 4, 2, 1 from left to right.

```
172 = 128 + 32 + 8 + 4  ->  10101100
192 = 128 + 64          ->  11000000
168 = 128 + 32 + 8      ->  10101000
130 = 128 + 2           ->  10000010
```

The subnet mask is n ones followed by zeros (32 - n of them). /24 means 24 ones, the same as 255.255.255.0. Where the mask has 1s, those bits are the network part. Where it has 0s, those bits are the host part.

The network address is the IP AND the mask:

```
192.168.1.130 AND 255.255.255.192
last octet: 10000010 AND 11000000 = 10000000 = 128
network = 192.168.1.128
```

Two special addresses exist in every subnet and can never be given to a device:

- Network address: all host bits 0 (names the subnet itself).
- Broadcast address: all host bits 1 (means everyone in the subnet).

Private ranges (use these for internal networks): 10.0.0.0/8, 172.16.0.0/12 (172.16.0.0 to 172.31.255.255), 192.168.0.0/16. Also avoid 127.0.0.0/8 (loopback) and 169.254.0.0/16 (link-local) for hosts. Anything else is public.

3. Subnets: the 5-step method

Given an IP and a prefix, find everything in 5 steps.

1. Find the interesting octet: the one where the mask is neither 255 nor 0. For /24, /16, /8 the boundary falls between octets, so the interesting octet is the last one of the network part.
2. Block size = 256 - mask value in that octet.
3. Network = the highest multiple of the block size that is less than or equal to the IP's value in that octet. Octets after it become 0.
4. Broadcast = network + block size - 1 in that octet. Octets after it become 255.
5. Usable range = network + 1 up to broadcast - 1. Host count = 2^(32 - n) - 2.

Example 1: 192.168.1.130/26. Mask octet 192, block 64. Multiples of 64: 0, 64, 128, 192. 130 falls in 128. Network 192.168.1.128, broadcast 192.168.1.191, usable .129 to .190, 62 hosts.

Example 2: 10.1.1.77/27. Mask octet 224, block 32. Multiples: 0, 32, 64, 96. 77 falls in 64. Network 10.1.1.64, broadcast 10.1.1.95, usable .65 to .94, 30 hosts.

Example 3 (big mask): 172.16.37.5/20. Mask is 255.255.240.0, so the interesting octet is the 3rd, block 16. 37 falls in 32. Network 172.16.32.0, broadcast 172.16.47.255, usable 172.16.32.1 to 172.16.47.254, 4094 hosts. Note that 172.16.32.255 is a valid host here, because the broadcast is 172.16.47.255.

Sizing a subnet: count every device that needs an address, including router interfaces, then find the smallest 2^h - 2 that fits. A LAN with 12 PCs and 1 router interface needs 13 addresses, so h = 4 (14 usable), which is a /28.

Point-to-point links between two routers: use a /30 (2 usable addresses). In NetPractice, treat /30 as the smallest sensible subnet. Do not use /31 or /32 for devices.

4. Devices: host, switch, router, Internet

Host: has an IP, a mask and a default gateway. It decides one thing: is the destination in my subnet (send directly) or not (send to the gateway)?

Switch: layer 2. It has no IP address in NetPractice and does not look at masks or routes. It just connects devices into one network (one broadcast domain). Everything attached to the same switch must be in the same subnet with the same mask.

Router: layer 3. It has two or more interfaces, and each interface has its own IP and mask in a different network. It forwards packets between networks using its routing table. The IP of a router interface is the gateway for the hosts on that interface's network.

Internet: in NetPractice it is just a far-away network with fixed values. Your job is to make the path toward it correct, usually with a default route.

MAC vs IP: a MAC address identifies a network card on the local link (layer 2). An IP address identifies a place in the network (layer 3). ARP is how a host finds the MAC for an IP on its own link: it broadcasts 'who has 10.0.1.1?' and the owner answers with its MAC. A host ARPs for the destination if it is local, or for the gateway if it is remote.

5. How a packet travels (both directions)

```
Host A 10.0.1.10/24 -- switch -- [R 10.0.1.1 | 10.0.2.1] -- switch -- Host B
10.0.2.10/24
```

A wants to ping B.

1. A checks: is 10.0.2.10 in my network? 10.0.2.10 AND 255.255.255.0 = 10.0.2.0, but my network is 10.0.1.0. No, so send to the default gateway 10.0.1.1.
2. The switch delivers the frame to R.
3. R looks up 10.0.2.10. The network 10.0.2.0/24 is directly connected on its second interface, so it delivers to B.
4. B replies to 10.0.1.10, which is not in its network, so it sends to its own gateway 10.0.2.1. R forwards it back to A.

Lessons: both hosts need a correct gateway, and the return path matters as much as the outgoing path. The source and destination IPs stay the same end to end, the MAC addresses change at every hop, and the TTL drops by 1 at every router.

6. Routing tables

Each row is Destination (a network with prefix) and Next hop (the IP of the next router's interface). Rules:

1. Networks directly connected to the router are known automatically. You only add routes for networks that are further away.
2. The next hop must be directly reachable, meaning it lies inside a subnet that one of the router's own interfaces belongs to.
3. The destination must be a proper network address with its prefix, for example 10.10.2.0/24. Writing 10.10.2.5/24 is wrong.
4. Longest prefix match wins. With routes 10.0.0.0/8 via X and 10.1.0.0/16 via Y, a packet for 10.1.2.3 goes via Y.
5. The default route is 0.0.0.0/0. It matches everything but loses to any more specific route. Use it to point toward the Internet or toward 'the rest of the network'.
6. Every router along the path needs a route to forward the packet, and the same is true for the reply. Always think about both directions.
7. Avoid loops: if R1's default goes to R2 and R2's default goes to R1, packets bounce until the TTL dies. Defaults should point toward the exit.
8. Aggregation: if several subnets form one block, one route can cover them, for example 10.10.0.0/16 via X covers 10.10.1.0/24 and 10.10.2.0/24. This is handy when the routing table has few rows.

7. Solving a level: the method

Do it in this order every time.

1. Read the whole level. Greyed-out fields are fixed constraints. Start from them and deduce the rest.
2. Draw the subnets. Every switch and every router-to-router link is one subnet. Every router interface belongs to exactly one subnet.
3. Decide the network and mask for each subnet. Same mask on every device in a subnet. No two subnets may overlap.
4. Assign IPs. Unique in the subnet, never the network or broadcast address, always inside the range.
5. Set each host's gateway to the router interface IP that sits in the host's own subnet.

6. Fill the routing tables: for every network a router is not directly connected to, add a route via the right next hop. Then check the reverse direction.
7. For anything toward the Internet, add the default route pointing to the next router or to the Internet interface.
8. Press Check again, and if something fails, read which device fails and trace the packet by hand instead of changing random values.

What the levels roughly test (they get harder, check your own):

- Early levels: one network, same mask, valid and unique IPs.
- Switch plus router: masks and gateways.
- Two or more networks: separate subnets, one router interface per network.
- Routing tables: next hops and default routes.
- Internet: default routes toward it.
- Last levels: several routers, subnet sizing, overlaps, and fixing a broken config with as few changes as possible.

Why a level fails, the usual suspects:

- A host uses the network or broadcast address.
- Masks differ between devices in the same subnet.
- The gateway is not inside the host's subnet, or it is the host's own IP.
- Duplicate IPs.
- Two interfaces of one router sit in the same or overlapping networks.
- The link between two routers has no subnet of its own.
- A missing return route.
- A next hop that is not directly connected.
- A route destination that is not a network address.
- Overlapping subnets in different parts of the topology.
- A routing loop between default routes.

8. Practice (do it on paper, answers at the end)

Part A: subnet math (network, broadcast, usable range, host count)

- A1. 192.168.10.200/26
- A2. 10.4.5.9/28

- A3. 172.20.100.130/25
- A4. 192.168.0.45/29
- A5. 10.200.77.5/18
- A6. Convert: 255.255.255.248 to CIDR, /21 to dotted, 255.255.240.0 to CIDR, /17 to dotted.

Part B: same network?

- B1. 192.168.1.70/26 and 192.168.1.120/26
- B2. 10.0.0.250/25 and 10.0.0.5/25
- B3. 172.16.5.1/23 and 172.16.4.200/23

Part C: valid host?

- C1. In 192.168.5.64/27, which are valid host addresses: .64, .70, .95, .96?
- C2. Is 10.0.0.255 valid as a host in 10.0.0.0/24? And in 10.0.0.0/23?

Part D: sizing

- D1. Smallest prefix for 25, 100, 2 and 500 hosts (counting only the hosts).
- D2. A LAN needs 12 PCs plus 1 router interface. Which prefix?

Part E: spot the bug

- E1. Host 10.0.0.70/26 with gateway 10.0.0.1.
- E2. Two hosts on one switch: 192.168.1.10/24 and 192.168.1.20/25.
- E3. A router with eth0 10.0.1.1/24 and eth1 10.0.1.129/25.
- E4. R1 route: destination 10.10.2.5/24 via 10.10.3.2.

Part F: build it

- F1. One router, two LANs. LAN1 has 50 hosts, LAN2 has 20 hosts. You own 192.168.1.0/24. Choose subnets, router IPs, host gateways, routes.
- F2. Two routers. LAN A 10.10.1.0/24 and LAN B 10.10.2.0/24, with R1 connected to R2 directly. Choose the link subnet, all IPs and all routes.
- F3. A chain: LAN 192.168.1.0/24, then R1, R2, R3, then the Internet. Use 10.0.12.0/30 for R1-R2, 10.0.23.0/30 for R2-R3 and 203.0.113.0/30 for R3-Internet. Give all IPs and routes so the LAN can reach the Internet and replies come back.

Answers

- A1. Block 64, 200 is in 192. Network 192.168.10.192, broadcast 192.168.10.255, usable .193 to .254, 62 hosts.
- A2. Block 16, 9 is in 0. Network 10.4.5.0, broadcast 10.4.5.15, usable .1 to .14, 14 hosts.
- A3. Block 128, 130 is in 128. Network 172.20.100.128, broadcast 172.20.100.255, usable .129 to .254, 126 hosts.
- A4. Block 8, 45 is in 40. Network 192.168.0.40, broadcast 192.168.0.47, usable .41 to .46, 6 hosts.
- A5. Mask 255.255.192.0, block 64 in the 3rd octet, 77 is in 64. Network 10.200.64.0, broadcast 10.200.127.255, usable 10.200.64.1 to 10.200.127.254, 16382 hosts.
- A6. /29, 255.255.248.0, /20, 255.255.128.0.
- B1. Yes. Both are in the 64 to 127 block, network 192.168.1.64.
- B2. No. 250 is in the 128 block, 5 is in the 0 block.
- B3. Yes. Mask 255.255.254.0, block 2 in the 3rd octet, both are in 172.16.4.0/23 (4.0 to 5.255).
- C1. Only .70. .64 is the network, .95 is the broadcast, .96 belongs to the next subnet.
- C2. In /24 it is the broadcast, so invalid. In /23 the range is 10.0.0.0 to 10.0.1.255, the broadcast is 10.0.1.255, so 10.0.0.255 is a valid host.
- D1. 25 hosts is /27 (30), 100 is /25 (126), 2 is /30 (2), 500 is /23 (510).
- D2. 13 addresses needed, 14 usable in a /28, so /28.
- E1. 10.0.0.70/26 is in 10.0.0.64/26 (64 to 127). The gateway 10.0.0.1 is in another subnet. It must be an address in this subnet, such as 10.0.0.65.
- E2. The masks differ (/24 vs /25), so the two hosts disagree about what is local. Use the same mask on both.
- E3. 10.0.1.129 is inside 10.0.1.0/24, so the interfaces overlap. Each interface must be in a different, non-overlapping network.
- E4. The destination must be the network address: 10.10.2.0/24.
- F1. LAN1 needs 50 + 1 = 51 addresses, so /26 (62). LAN2 needs 20 + 1 = 21, so /27 (30). LAN1 = 192.168.1.0/26 (usable .1 to .62), LAN2 = 192.168.1.64/27 (usable .65 to .94). Router eth0 = 192.168.1.1/26, eth1 = 192.168.1.65/27. LAN1 hosts use .2 to .62 with gateway 192.168.1.1. LAN2 hosts use .66 to .94 with gateway 192.168.1.65. No routes are needed because both networks are directly connected.

- F2. Link subnet 10.10.3.0/30. R1 has 10.10.1.1/24 and 10.10.3.1/30. R2 has 10.10.3.2/30 and 10.10.2.1/24. Hosts in A use gateway 10.10.1.1, hosts in B use 10.10.2.1. Routes: R1 sends 10.10.2.0/24 via 10.10.3.2, R2 sends 10.10.1.0/24 via 10.10.3.1.
- F3. R1: 192.168.1.1/24 and 10.0.12.1/30. R2: 10.0.12.2/30 and 10.0.23.1/30. R3: 10.0.23.2/30 and 203.0.113.1/30. Internet side is 203.0.113.2. LAN hosts use gateway 192.168.1.1. Routes: R1 default via 10.0.12.2. R2 192.168.1.0/24 via 10.0.12.1, default via 10.0.23.2. R3 192.168.1.0/24 via 10.0.23.1, default via 203.0.113.2. The LAN route on R2 and R3 is what makes the reply come back. In real life you would also need NAT on R3, because private addresses are not routed on the Internet.

9. Evaluation prep

What to expect (typical, check your eval scale): the evaluator loads your submitted config and checks that the levels pass, then asks you to explain some levels and may ask you to solve or change one live. Cheating shows immediately, because you cannot explain a config you did not build. You must be able to redo every level yourself.

Likely questions, with short model answers:

- What is an IP address? A 32-bit identifier for a device's place in a network, written as 4 octets.
- What is a subnet mask? It marks which bits are the network part and which are the host part. It lets a device decide if another IP is local or needs a router.
- What is CIDR notation? Writing the mask as the number of 1 bits, like /24, instead of 255.255.255.0.
- Why can't I use the first and last address of a subnet? The first is the network address and the last is the broadcast address, so they are reserved. That is why usable hosts = 2^h - 2.
- Switch vs router? A switch (layer 2) connects devices in the same network and does not look at IPs. A router (layer 3) connects different networks and forwards packets using a routing table.
- What is a default gateway? The router interface a host sends traffic to when the destination is outside its own subnet. It must be in the host's subnet.
- What is a routing table? A list of destination networks and the next hop to reach them. The router picks the most specific match.
- What is the default route? 0.0.0.0/0. It matches everything and is used when no more specific route exists, typically toward the Internet.

- Why do I need routes in both directions? Replies are packets too. If the return path is missing, the ping gets there but the answer never comes back.
- Why can't two interfaces of a router be in the same network? The router would not know which interface leads to which side. Each interface must be in its own non-overlapping network.
- What happens with duplicate IPs? Both devices claim the same address, so traffic goes to the wrong one or to both. It breaks communication.
- Public vs private IPs? Private ranges (10/8, 172.16/12, 192.168/16) are for internal networks and are not routed on the Internet. Public ones are globally unique. NAT lets private hosts reach the Internet through a public address.
- What is ARP? The protocol that finds the MAC address for an IP on the local link.
- IPv4 vs IPv6? IPv4 is 32 bits, IPv6 is 128 bits, written in hex groups. IPv6 was created because IPv4 addresses ran out.
- How many hosts are in a /29? 2^3 - 2 = 6.

Tips for the defense:

- Explain each level with the same method: subnets first, then IPs, gateways, routes.
- Do the subnet math out loud with the block-size trick. It shows you understand it.
- If asked to fix a broken level, find the cause by tracing the packet in both directions instead of guessing.
- Practice solving 2 or 3 levels from scratch, with a timer, right before the eval.

10. Final checklist before you submit

- Every level shows success in the interface, and you pressed Check again after your last change.
- You downloaded the config exactly as the subject says and used the right filename and location.
- No device uses a network or broadcast address.
- Masks match inside every subnet, and no subnets overlap.
- All IPs are unique inside their subnet.
- Every host gateway is a router interface in the host's own subnet.
- Every route has a proper network destination and a directly reachable next hop, and both directions are covered.
- You can redo any level from scratch and explain every field out loud.